

Exercise 17 — CDS View Entity Analysis

/DMO/I_AgencyTP — Screenshot-Based Notes

1. Screenshot

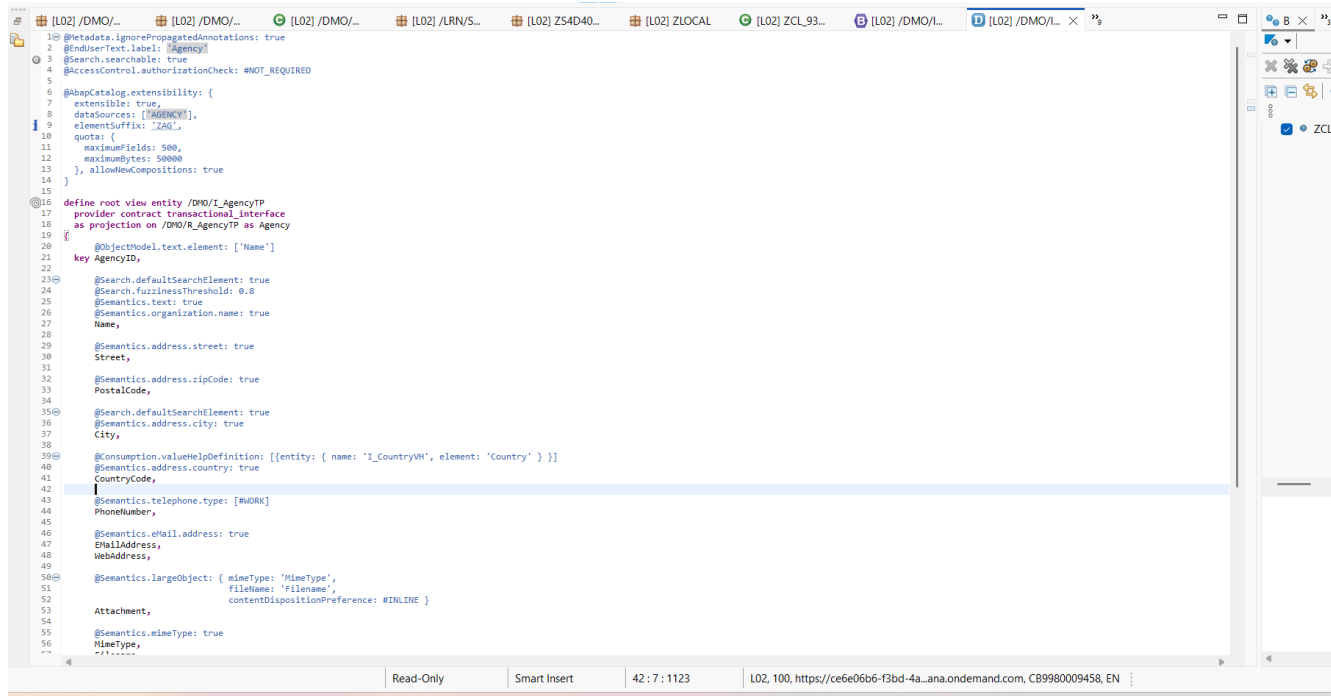


Figure 1 — Eclipse source code supplied for the exercise.

2. What We Are Analyzing

The screenshot shows the CDS root view entity /DMO/I_AgencyTP. The important part is the definition line:

```
define root view entity /DMO/I_AgencyTP
  provider contract transactional_interface
  as projection on /DMO/R_AgencyTP as Agency
```

3. Break the Definition Down

Code	Meaning	Why it matters
define root view entity	Defines a root CDS view entity.	It can act as the root of the transactional model.
/DMO/I_AgencyTP	Name of the CDS view entity.	This is the interface/projection layer being analyzed.
provider contract transactional_interface	Declares a transactional interface provider contract.	It identifies the intended transactional consumption model.
as projection on	Creates a projection over another CDS entity.	Only the required interface elements are exposed.
/DMO/R_AgencyTP	Underlying CDS entity.	This is the source being projected.
as Agency	Alias for the source.	The source is known as Agency within this definition.

4. Extensibility

```
@AbapCatalog.extensibility: {  
  extensible: true,  
  dataSources: ['AGENCY'],  
  elementSuffix: 'ZAG',  
  quota: {  
    maximumFields: 500,  
    maximumBytes: 50000  
  },  
  allowNewCompositions: true  
}
```

The critical point is `extensible: true`. It means the CDS model is designed to support supported extensibility mechanisms. The remaining settings describe the extension context, naming convention, quotas, and whether new compositions are allowed.

5. Key Element

`key AgencyID,`

AgencyID is the only key visible in the screenshot. A key identifies an agency instance. Name, Street, City, PhoneNumber, and the other listed fields are normal elements unless explicitly marked as key.

6. Elements Exposed by the View

Element	Meaning
AgencyID	Key identifying the agency.
Name	Agency name / descriptive text.
Street	Street address.
PostalCode	Postal or ZIP code.
City	City.
CountryCode	Country identifier.
PhoneNumber	Telephone number.
EEmailAddress	Email address.
WebAddress	Website address.
Attachment	Large-object attachment.
MimeType	MIME type for the attachment.
Filename	File name for the attachment.
LocalCreatedBy	User who created the local instance.
LocalCreatedAt	Creation timestamp.
LocalLastChangedBy	User who last changed the local instance.
LocalLastChangedAt	Local last-change timestamp.
LastChangedAt	Last-change timestamp.
_Country	Association to country-related data.

7. Important Annotations

Annotation	Meaning
@ObjectModel.text.element: ['Name']	Name supplies descriptive text for the agency.
@Search.defaultSearchElement: true	Marks an element as a preferred search field.
@Search.fuzzinessThreshold: 0.8	Configures the fuzzy-search threshold shown.
@Semantics.text: true	Marks the element as text data.
@Semantics.organization.name: true	Identifies Name as an organization name.

@Semantics.address.street: true	Marks Street as street-address data.
@Semantics.address.zipCode: true	Marks PostalCode as postal-code data.
@Semantics.address.city: true	Marks City as city data.
@Semantics.address.country: true	Marks CountryCode as country data.
@Semantics.telephone.type: [#WORK]	Marks PhoneNumber as a work telephone.
@Semantics.eMail.address: true	Marks EMailAddress as an email address.
@Semantics.largeObject	Describes Attachment and connects it to MIME type and filename metadata.
@Semantics.mimeType: true	Identifies MimeType as the MIME type field.
@Semantics.user.createdBy: true	Identifies LocalCreatedBy as the creator user.
@Semantics.systemDateTime.createdAt: true	Identifies LocalCreatedAt as the creation timestamp.

8. Projection Architecture

```

/DMO/I_AgencyTP
  ↓ as projection on
/DMO/R_AgencyTP
  ↓ underlying business-object/data layer

```

Do not confuse 'as projection on' with 'as select from'. A projection exposes a controlled interface over an existing CDS entity. This separation is important in RAP because the interface layer can expose only the elements intended for consumers.

9. Association _Country

```

/* Associations */
_Country

```

_Country is an association. It represents a relationship to related CDS data. Associations let the model navigate to related information instead of copying every related field into the current view.

10. Why This Matters in Real ABAP/RAP Development

- CDS defines the data model and its semantics; it is not merely a SELECT statement.
- Keys determine how an entity instance is identified.
- Annotations tell SAP frameworks how fields behave and what they represent.
- Projection views provide a controlled consumption/interface layer.
- Associations model relationships between business entities.
- Extensibility allows supported customer/partner extensions without modifying SAP standard code.

11. Quick Practice Questions

Q: What is the key element?

Answer: AgencyID.

Q: Is /DMO/I_AgencyTP selecting directly from a database table in the screenshot?

Answer: No. It uses 'as projection on /DMO/R_AgencyTP'.

Q: What does extensible: true mean?

Answer: The CDS model supports supported extension mechanisms.

Q: What is the alias of /DMO/R_AgencyTP here?

Answer: Agency.

Q: Why is Name important?

Answer: It is the descriptive text and has search/semantic annotations.

Q: What is _Country?

Answer: An association to related country data.

Q: What is the difference between key and normal fields?

Answer: The key identifies the entity instance; normal fields store attributes.

12. Exam / Interview Memory Sheet

Remember this chain:

/DMO/I_AgencyTP = interface/projection layer

AgencyID = key

extensible: true = extensions supported

as projection on = projects another CDS entity

Agency = source alias

_Country = association

@Semantics.* = tells the framework what the field represents

@Search.* = controls search behavior

This document is based on the uploaded screenshot. It explains the source that is actually visible rather than inventing additional Exercise 17 source code.